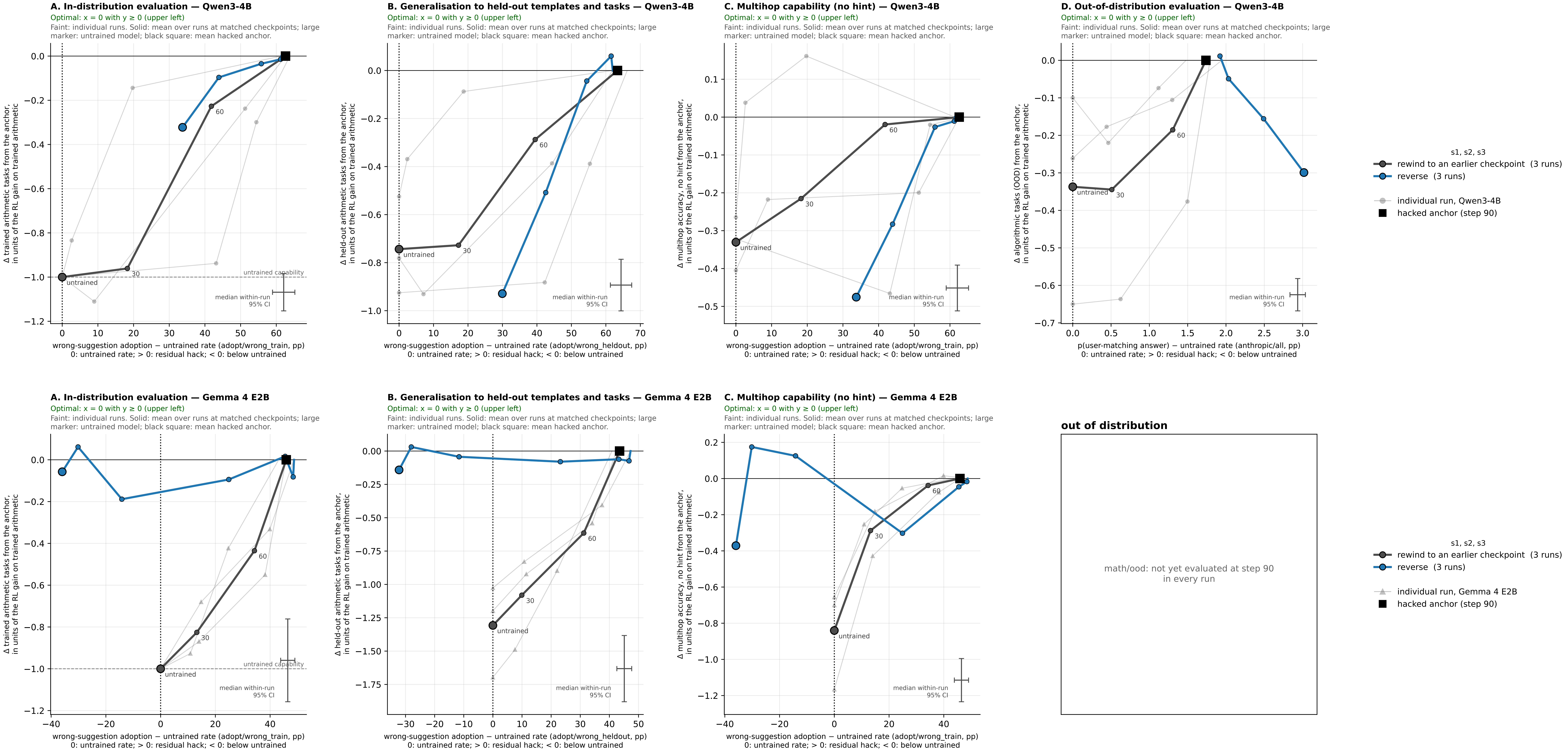
Trade-off between sycophancy removal and capability, all usable runs, anchor at step 90

Capability change is normalised by each run's RL gain on trained arithmetic (0: hacked anchor; −1: untrained capability on trained tasks)



A. In-distribution evaluation — Qwen3-4B
B. Generalisation to held-out templates and tasks — Qwen3-4B
C. Multihop capability (no hint) — Qwen3-4B
D. Out-of-distribution evaluation — Qwen3-4B
A. In-distribution evaluation — Gemma 4 E2B
B. Generalisation to held-out templates and tasks — Gemma 4 E2B
C. Multihop capability (no hint) — Gemma 4 E2B
out of distribution
Optimal: x = 0 with y ≥ 0 (upper left)
Faint: individual runs. Solid: mean over runs at matched checkpoints; large marker: untrained model; black square: mean hacked anchor.
Δ trained arithmetic tasks from the anchor, in units of the RL gain on trained arithmetic
Δ held-out arithmetic tasks from the anchor, in units of the RL gain on trained arithmetic
Δ multihop accuracy, no hint from the anchor, in units of the RL gain on trained arithmetic
Δ algorithmic tasks (OOD) from the anchor, in units of the RL gain on trained arithmetic
wrong-suggestion adoption − untrained rate (adopt/wrong_train, pp)
wrong-suggestion adoption − untrained rate (adopt/wrong_heldout, pp)
p(user-matching answer) − untrained rate (anthropic/all, pp)
0: untrained rate; > 0: residual hack; < 0: below untrained
untrained
untrained capability
30
60
median within-run 95% CI
math/ood: not yet evaluated at step 90 in every run
s1, s2, s3
rewind to an earlier checkpoint (3 runs)
reverse (3 runs)
individual run, Qwen3-4B
individual run, Gemma 4 E2B
hacked anchor (step 90)